

CEN 352 – Project Proposal

TAIRA: Traffic Accident Intelligent Risk Advisor

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Course: CEN 352 – Artificial Intelligence

Project Type: Hybrid Intelligent Agent (Machine Learning + Rule-Based System)

1. Introduction & Motivation

Traffic accidents remain a major urban safety concern, influenced by road conditions, weather, lighting, and driver behavior. Cities collect large crash datasets, but these are rarely turned into tools that help users **interpret risks** or **explore how different conditions affect severity**.

This project proposes **TAIRA – Traffic Accident Intelligent Risk Advisor**, a hybrid AI agent that:

- Predicts the **Most Severe Injury** of a crash using machine learning.
- Applies **rule-based reasoning** to generate an interpretable **risk level** (LOW / MEDIUM / HIGH / CRITICAL).
- Provides a **Streamlit web interface** to let users test scenarios and view crash patterns.

The system is meant as an educational, non-real-time, decision-support agent demonstrating

2. Problem Definition (PEAS Framework)

Performance

- Predict crash injury severity with high **Accuracy** and **Weighted F1-score**.
- Provide meaningful risk levels and short explanations for decisions.
- Present clear visualizations of underlying crash patterns.

Environment

- Historical urban traffic accident records (Chicago dataset).
- Each record includes: weather, lighting, road surface, traffic control device, crash type, and injury outcomes.

Actuators

- Predicted injury severity.
- Risk level (text output).
- Generated explanation.
- Dashboard: charts for month/hour/day, weather, lighting, crash types.

Sensors

- User-selected dropdown inputs (traffic control device, weather, lighting, road surface, cause, crash type).
 - Encoded dataset features used by the ML model.
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3. Techniques Used (Hybrid Agent)

A. Machine Learning – Random Forest Classifier

- Predicts the `MOST_SEVERE_INJURY` category.
- Uses only categorical features (one-hot encoded).
- Random Forest chosen for:
 - Strong performance on structured categorical data.
 - Robustness to noise and non-linear interactions.
- Evaluated using Accuracy and Weighted F1-score.

B. Rule-Based Reasoning – Expert System Layer

After prediction, deterministic rules evaluate environmental danger:

Examples:

- If ML predicts *Fatal* or *Incapacitating* → **CRITICAL** risk.
- If **SNOW/ICE + DARKNESS + NO CONTROLS** → high risk regardless of ML output.
- If mildly adverse conditions (wet road + night + turning crash) → **MEDIUM**.
- Otherwise → **LOW**.

Rules also produce a short explanation (e.g., “Wet surface and no traffic control increase danger”).

C. Hybrid Agent Structure

The `TrafficAccidentAgent`:

1. Accepts user selections.
2. Uses Random Forest to predict injury severity.
3. Applies rule-based logic to produce risk level.
4. Returns results to the UI (prediction + risk + explanation).

This satisfies the “Integration of Multiple AI Techniques” requirement.

4. Dataset Summary

Using the **Traffic Crashes Dataset**, which includes:

- TRAFFIC_CONTROL_DEVICE
- WEATHER_CONDITION
- LIGHTING_CONDITION
- ROADWAY_SURFACE_COND
- FIRST_CRASH_TYPE
- PRIM_CONTRIBUTORY_CAUSE
- MOST_SEVERE_INJURY (target label)

The dataset contains tens of thousands of records—enough to train a meaningful classifier.

6. Originality

The project is original because:

- It integrates **two AI paradigms** (ML + rule-based logic) into a single agent, not just a simple model.
- It focuses on **pre-crash scenario analysis**, not mere dataset description.
- It builds a professional, multi-page **web dashboard** that behaves like a real tool.
- It provides **explanations**, which most student ML projects do not include.

This goes beyond typical Kaggle-style work and demonstrates concepts from the CEN 352 course.

7. Ethical Considerations

- The model may reflect dataset biases (location, time of day, underreported injuries).
 - Users may overtrust the system → **automation bias**.
 - TAIRA is **not** for real-world safety decisions; it is a learning tool.
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